

Chapter 4: Linear Dynamics

T. K. Deskins, Ph.D.

Professor of Physics & Chemistry

Southern Union State Community College

PHY201
Lecture 4



Part I

Forces and Newton's Laws of Motion

Outline of Part I: Forces and Newton's Laws of Motion

Introduction to Dynamics

Newton's First Law — Inertia

Newton's Second Law of Motion

Newton's Third Law of Motion

Kinematics vs. Dynamics

Kinematics (Ch. 2–3)

Describes **how** objects move.

- ▶ Position, velocity, acceleration
- ▶ Does not ask *why*

Dynamics (Ch. 4)

Explains **why** objects move.

- ▶ Forces cause changes in motion
- ▶ Asks: *what force produces this $\vec{a}$?*

Cause and Effect

Effect ← **Cause**

Rain ← Storm clouds

Sunburn ← The Sun

Acceleration ← **Net Force**

Dynamics: the study of how forces affect the motion of objects.

Force

Force ($\vec{F}$) — SI Unit: Newton (N)

A **force** is a push or a pull on an object.

Force is a **vector** quantity — it has both magnitude and direction.

Examples of Forces

- ▶ Pushing a door open
- ▶ Pulling a rope
- ▶ Gravity pulling you toward Earth
- ▶ A surface pushing back on your feet

Multiple forces on one object add to give a single **net force**:

$$\vec{F}_{\text{net}} = \sum \vec{F}$$

Because force is a vector, forces combine by **vector addition**.

Outline of Part I: Forces and Newton's Laws of Motion

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Newton's First Law of Motion

Newton's First Law — The Law of Inertia

A body at rest remains at rest, or, if in motion, remains in motion at a **constant velocity** unless acted on by a **net external force**.

Inertia

Inertia is the tendency of an object to resist changes in its state of motion.

- ▶ An object at rest **stays at rest** unless pushed or pulled.
- ▶ A moving object **keeps moving** at the same speed and direction unless a net force acts.
- ▶ **Constant velocity (including rest) is the natural state.**

The First Law reframes the question:

Instead of “*why do moving objects stop?*”

ask “*what force is causing the deceleration?*”

Mass vs. Weight

Mass (m) — SI Unit: kg

- ▶ Quantity of matter in an object
- ▶ Measure of **inertia**
- ▶ **Invariant**: does not change with location

Weight ($\vec{w}$) — SI Unit: N

- ▶ Gravitational force on an object
- ▶ **Depends on location** (different on Earth vs. Moon)
- ▶ Directed **downward** toward Earth's center

Mass and weight are **not** the same thing.
Mass is measured in kg; weight is measured in N.

A 1 kg object weighs $w = mg = (1 \text{ kg})(9.80 \frac{\text{m}}{\text{s}^2}) = 9.80 \text{ N}$ on Earth, but only 1.63 N on the Moon.

Check Your Understanding

Which has more mass — a kilogram of cotton balls or a kilogram of gold?

Come to lecture to see the answer.

Outline of Part I: Forces and Newton's Laws of Motion

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Newton's Second Law of Motion

Newton's Second Law

The acceleration of a system is **directly proportional to** and in the **same direction** as the net external force, and **inversely proportional** to its mass:

$$\vec{F}_{\text{net}} = m\vec{a}$$

where $\vec{F}_{\text{net}}$ is the net external force (N), m is the mass (kg), and $\vec{a}$ is the acceleration ($\frac{\text{m}}{\text{s}^2}$).

Solved for acceleration:

$$\vec{a}_x = \frac{1}{m} \vec{F}_{\text{net},x}$$

- ▶ $\vec{a} \propto \vec{F}_{\text{net}}$: double force $\Rightarrow$ double acceleration
- ▶ $\vec{a} \propto \frac{1}{m}$: double mass $\Rightarrow$ half acceleration
- ▶ $\vec{a}$ is in the **same direction** as $\vec{F}_{\text{net}}$

Net Force ($\vec{F}_{\text{net}}$)

The **net force** is the **vector sum** of all external forces acting on an object:

$$\vec{F}_{\text{net},x} = \sum \vec{F}_x$$

Analogy: Net vs. Gross

► You earn \$10/hr (*gross*), taxes take \$1/hr, you receive \$9/hr (*net*)

Likewise: $\vec{F}_{\text{push}} = 10 \text{ N}$ and friction $\vec{f} = -1 \text{ N}$, so:

$$\vec{F}_{\text{net}} = 10 \text{ N} + (-1 \text{ N}) = \boxed{9 \text{ N}}$$

Only **external** forces contribute to $\vec{F}_{\text{net}}$. Internal forces between parts of a system cancel.

The Newton (Unit)

Definition of the Newton (N)

$$1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$$

One newton is the force required to give a 1 kg object an acceleration of $1 \frac{\text{m}}{\text{s}^2}$.

- ▶ A small apple weighs approximately 1 N
- ▶ A 150-lb person weighs approximately 667 N
- ▶ $1 \text{ N} \approx 0.225 \text{ lb}$

$$\vec{F}_{\text{net},x} = m\vec{a}_x \quad \Longleftrightarrow \quad \text{N} = \text{kg} \cdot \frac{\text{m}}{\text{s}^2}$$

Weight

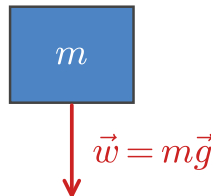
Weight ($\vec{w}$) — SI Unit: N

Weight is the **gravitational force** on an object of mass m :

$$\vec{w} = m\vec{g}$$

where m is the object's mass and $g = 9.80 \frac{\text{m}}{\text{s}^2}$ is the magnitude of the gravitational acceleration at Earth's surface. Weight is directed **downward**.

Weight is a **force** (vector, in N).
Mass is a **scalar** (in kg).



Example: Force on a Crate

Jimmy the Monkey Pushes a Crate — (cp2e_ch4_ex3)

Jimmy the monkey pushes a 10.0 kg crate with a force of 100 N.
The friction force acting on the crate is 15.0 N.
Find the magnitude of the crate's acceleration.

$$m = 10.0 \text{ kg}, \vec{F}_p = 100 \text{ N}, \vec{f} = -15.0 \text{ N}$$

$$\vec{F}_{\text{net},x} = \sum \vec{F}_x = \vec{F}_p + \vec{f}$$

$$\vec{F}_{\text{net},x} = 100 \text{ N} + (-15.0 \text{ N}) = 85.0 \text{ N}$$

$$\vec{a}_x = \frac{\vec{F}_{\text{net},x}}{m} = \frac{85.0 \text{ N}}{10.0 \text{ kg}}$$

$$\boxed{\vec{a}_x = 8.50 \frac{\text{m}}{\text{s}^2}}$$

Friction opposes the direction of motion, so it is **negative** in our coordinate system.

- ▶ Always define positive direction first.
- ▶ Forces opposing motion are negative.

Check Your Understanding

Which statement is correct?

(a) Net force causes **motion**.

(b) Net force causes **change in motion**.

Explain your answer and give an example.

Come to lecture to see the answer.

Check Your Understanding

A rock is thrown straight up.

What is the net external force acting on the rock at the very **top** of its trajectory?

Come to lecture to see the answer.

Outline of Part I: Forces and Newton's Laws of Motion

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Newton's Third Law

Whenever one body exerts a force on a second body, the first body experiences a force that is **equal in magnitude** and **opposite in direction**.

Action–Reaction Pairs

Action force	Reaction force
Swimmer pushes wall backward	Wall pushes swimmer forward
Foot pushes ground backward	Ground pushes foot forward
Rocket expels gas backward	Gas pushes rocket forward

Thrust: the reaction force that propels rockets, planes, and cars forward.

Newton's Third Law — Common Misconception

Equal-and-opposite forces **do not cancel**.

They act on **different objects** — they are never part of the same free-body diagram.

- ▶ Earth pulls **you** downward (your weight).
- ▶ You pull **Earth** upward with equal force.
- ▶ These forces act on **different objects**, so they do not cancel.
- ▶ To analyze *your* motion, you use only the forces on *you*.

When analyzing a system, define the system boundary first. Forces crossing the boundary are **external**; forces entirely within the system are **internal** and cancel.

Check Your Understanding

When you take off in a jet aircraft, there is a sensation of being pushed back into your seat.

Is there actually a force pushing **backward** on you? Explain using Newton's Third Law.

Come to lecture to see the answer.

Part II

Free-Body Diagrams and Types of Forces

Outline of Part II: Free-Body Diagrams and Types of Forces

Free-Body Diagrams

Free-Body Diagrams

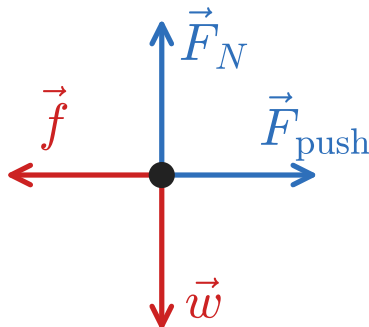
Free-Body Diagram (FBD)

A **free-body diagram** is a sketch in which:

- ▶ the object of interest is represented as a **single dot**
- ▶ all **external forces** on that object are drawn as arrows from the dot
- ▶ internal forces are excluded

Drawing an FBD

1. Draw a dot representing the object.
2. Identify **every** external force.
3. Draw each force as an arrow from the dot.
4. Label each arrow.



Normal Force and Weight

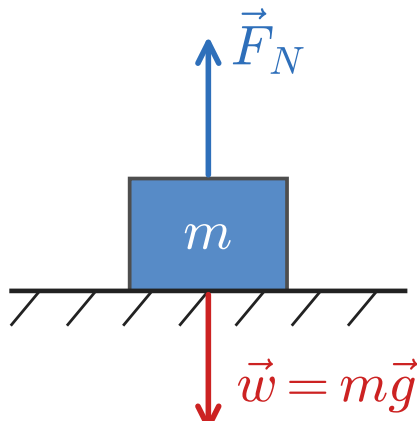
Normal Force ($\vec{F}_N$) – SI Unit: N

The **normal force** is the contact force a surface exerts on an object, directed **perpendicular** to the surface.

- ▶ “Normal” means perpendicular (geometry term)
- ▶ Purpose: prevent objects from passing through each other
- ▶ On a flat, horizontal surface:
 $\vec{F}_N = -\vec{w} = m\vec{g}$ (when $a_y = 0$)

$\vec{F}_N \neq m\vec{g}$ in general. This holds **only** on a flat surface with no vertical acceleration.

$$\vec{a}_y = 0 \because \Sigma \vec{F}_y = 0$$



Tension and Friction

Tension ($\vec{T}$) — SI Unit: N

Tension is the pulling force transmitted along a flexible connector (rope, cable, string).

- ▶ Tension always **pulls** — it cannot push.
- ▶ Acts along the connector, away from the object.

You cannot push a rope.

Friction ($\vec{f}$) — SI Unit: N

Friction is a contact force between surfaces that opposes relative sliding motion.

- ▶ Always acts **opposite to the direction of motion** (or attempted motion)
- ▶ Friction is a real external force

Part III

Solving Newton's Second Law Problems

Outline of Part III: Solving Newton's Second Law Problems

Problem-Solving Strategy

Inclined Planes

Connected Systems and Tension

How to Solve a Dynamics Problem

Four-Step Strategy

1. **Draw a physical representation** of the situation.
2. **Draw a free-body diagram** (FBD) for each object of interest.
3. **Apply Newton's Second Law** in all relevant directions:

$$\sum \vec{F}_x = m\vec{a}_x \quad \sum \vec{F}_y = m\vec{a}_y$$

(or use $\parallel$ and $\perp$ axes when working on an inclined plane)

4. **Solve** the resulting equations for the unknown(s).

 Check your answer: do the units work out? Is the magnitude reasonable? Does the direction make sense?

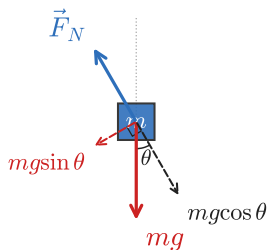
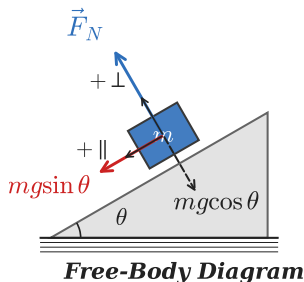
Outline of Part III: Solving Newton's Second Law Problems

Problem-Solving Strategy

Inclined Planes

Connected Systems and Tension

Forces on an Inclined Plane



On an inclined plane at angle θ , it is convenient to choose axes **parallel** ($\parallel$) and **perpendicular** ($\perp$) to the slope.

Weight $\vec{w} = m\vec{g}$ has two components:

$$W_{\parallel} = mg \sin \theta \quad (\text{along slope})$$

$$W_{\perp} = mg \cos \theta \quad (\text{into surface})$$

The normal force equals the perpendicular component of weight (when $a_{\perp} = 0$):

$$\vec{F}_N = mg \cos \theta$$

Acceleration on a Frictionless Inclined Plane

Applying $\sum \vec{F}_{\parallel} = m\vec{a}_{\parallel}$ along the slope (taking down-slope as positive):

$$W_{\parallel} = m\vec{a}_{\parallel}$$

$$mg \sin \theta = m\vec{a}_{\parallel}$$

$$\vec{a}_{\parallel} = g \sin \theta$$

where $\vec{a}_{\parallel}$ is the acceleration along the slope, $g = 9.80 \frac{\text{m}}{\text{s}^2}$, and θ is the angle of the slope from horizontal.

Mass **cancels out** — all objects accelerate at the same rate down a frictionless incline regardless of mass.

Example: Snowboarder on a Slope

Anna Snowboards Down a Slope — (cp2e_ch4)

Anna ($m = 60.0 \text{ kg}$) snowboards down a slope inclined at $\theta = 10.0^\circ$. Assume the slope is frictionless.

Find (a) her acceleration down the slope and (b) the normal force.

$$m = 60.0 \text{ kg}, \theta = 10.0^\circ, g = 9.80 \frac{\text{m}}{\text{s}^2}$$

$$(a) \quad \sum \vec{F}_{\parallel} = m\vec{a}_{\parallel}$$

$$mg \sin \theta = m\vec{a}_{\parallel}$$

$$\vec{a}_{\parallel} = g \sin 10.0^\circ = 1.70 \frac{\text{m}}{\text{s}^2}$$

Note: we used **two independent equations**, one for each direction.

The $\parallel$ equation gives $\vec{a}$; the $\perp$ equation gives $\vec{F}_N$.

- Mass cancels in part (a) — same $\vec{a}$ for any mass.
- $\vec{F}_N < mg$ on a slope ($\cos \theta < 1$).

Example: Snowboarder on a Slope

Anna Snowboards Down a Slope — (cp2e_ch4)

Anna ($m = 60.0 \text{ kg}$) snowboards down a slope inclined at $\theta = 10.0^\circ$. Assume the slope is frictionless.

Find (a) her acceleration down the slope and (b) the normal force.

$$m = 60.0 \text{ kg}, \theta = 10.0^\circ, g = 9.80 \frac{\text{m}}{\text{s}^2}$$

$$(b) \quad \sum \vec{F}_\perp = 0 \quad \because a_\perp = 0$$

$$\vec{F}_N - mg \cos \theta = 0$$

$$\boxed{\vec{F}_N = mg \cos 10.0^\circ = 578 \text{ N}}$$

Note: we used **two independent equations**, one for each direction.

The $\parallel$ equation gives $\vec{a}$; the $\perp$ equation gives $\vec{F}_N$.

- Mass cancels in part (a) — same $\vec{a}$ for any mass.
- $\vec{F}_N < mg$ on a slope ($\cos \theta < 1$).

Check Your Understanding

A block slides down a frictionless ramp inclined at 25.0° .

- (a) What is its acceleration?
- (b) Does the answer depend on the block's mass?

Come to lecture to see the answer.

Outline of Part III: Solving Newton's Second Law Problems

Problem-Solving Strategy

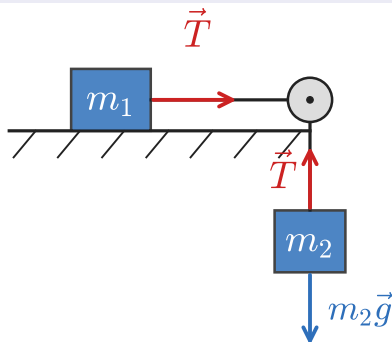
Inclined Planes

Connected Systems and Tension

Objects Connected by a Rope

When two objects are connected by a **massless, inextensible** rope over a **frictionless pulley**:

- ▶ The tension T is the **same throughout** the rope.
- ▶ Both objects have the **same magnitude of acceleration**.



Strategy: draw a separate FBD for each object. Write $\sum \vec{F} = m\vec{a}$ for each. Use the shared T and a to link the equations.

- ▶ $T_1 = T_2 = T$ (massless rope)
- ▶ $a_1 = a_2 = a$ (inextensible rope)

Example: Modified Atwood Machine

Block on a Surface Connected to a Hanging Mass — (cp2e_ch4)

A 4.00 kg block sits on a frictionless horizontal surface, connected by a rope over a pulley to a hanging 3.00 kg mass.

Find (a) the acceleration of the system and (b) the tension in the rope.

$$m_1 = 4.00 \text{ kg}, m_2 = 3.00 \text{ kg}$$

$$\text{For } m_1 (x) : T = m_1 a$$

$$\text{For } m_2 (y) : m_2 g - T = m_2 a$$

$$\text{Add equations: } m_2 g = (m_1 + m_2) a$$

$$a = \frac{m_2 g}{m_1 + m_2} = \frac{(3.00)(9.80)}{7.00}$$

$$a = 4.20 \frac{\text{m}}{\text{s}^2}$$

$$T = m_1 a = (4.00)(4.20) = 16.8 \text{ N}$$

Check:

$$T = m_2(g - a) = 3.00(9.80 - 4.20) = 3.00(5.60) = 16.8 \text{ N } \checkmark$$

- Treat each mass separately.
- $T < m_2 g$ because m_2 is **accelerating** — not in equilibrium.

Part IV

Further Applications

Outline of Part IV: Further Applications

Drag Force and Terminal Velocity

The Four Basic Forces

Drag Force

Drag Force ($\vec{F}_D$) — SI Unit: N

The **drag force** is the frictional force exerted by a **fluid** (air or water) on an object moving through it. It opposes the direction of motion:

$$F_D = \frac{1}{2} C \rho A v^2$$

where C is the dimensionless drag coefficient, ρ is the fluid density ($\frac{\text{kg}}{\text{m}^3}$), A is the cross-sectional area (m^2), and v is the speed relative to the fluid.

- ▶ For air at 15°C: $\rho \approx 1.225 \frac{\text{kg}}{\text{m}^3}$
- ▶ C is determined by experiment. This will be given to you for a problem.
- ▶ $F_D \propto v^2$ — drag grows rapidly with speed

Terminal Velocity

Terminal Velocity (v_e)

As an object falls and speeds up, drag increases. **Terminal velocity** is reached when drag equals weight and acceleration becomes zero:

$$\vec{F}_D + \vec{w} = 0 \quad \Rightarrow \quad F_D = w$$

$$\frac{1}{2} C \rho A v_e^2 = mg$$

$$v_e = \sqrt{\frac{2mg}{C\rho A}}$$

where v_e is the terminal (equilibrium) velocity and all other variables are as defined above.

At terminal velocity: $a_y = 0$ and $v = \text{constant}$.

The object is in **dynamic equilibrium** in the vertical direction.

Example: Terminal Velocity of a Falling Squirrel

Terminal Velocity — (cp2e_ch4)

A squirrel falls from a tree. Its body acts like a parachute with dimensions $L = 30$ cm, $W = 25$ cm. Its mass is $m = 400$ g. Take $C = 1$, $\rho_{\text{air}} = 1.225 \frac{\text{kg}}{\text{m}^3}$. Find the terminal velocity.

$$A = LW = (0.30 \text{ m})(0.25 \text{ m}) = 0.075 \text{ m}^2$$

$$m = 0.400 \text{ kg}$$

$$v_e = \sqrt{\frac{2mg}{C\rho A}} = \sqrt{\frac{2(0.400)(9.80)}{(1)(1.225)(0.075)}}$$

$$v_e = \sqrt{\frac{7.84}{0.0919}} = \sqrt{85.3}$$

$$v_e = 9.24 \frac{\text{m}}{\text{s}} \approx 9.2 \frac{\text{m}}{\text{s}}$$

$$9.2 \frac{\text{m}}{\text{s}} \approx 21 \frac{\text{m}}{\text{h}}$$

Squirrels are well-designed for this: large surface area relative to body mass lowers terminal velocity. They survive such falls routinely.

- Larger $A \Rightarrow$ lower v_e (parachute effect)
- Larger $m \Rightarrow$ higher v_e

Outline of Part IV: Further Applications

Drag Force and Terminal Velocity

The Four Basic Forces

The Four Fundamental Forces of Nature

All forces in nature reduce to exactly **four fundamental forces**:

Force	Rel. Strength	Range	Everyday Role
Strong nuclear	1	$\sim 10^{-15}$ m	Holds nucleus together
Electromagnetic	10^{-2}	Infinite	Contact, friction, tension
Weak nuclear	10^{-6}	$< 10^{-18}$ m	Radioactive decay
Gravitational	10^{-38}	Infinite	Weight, orbits

- ▶ Nearly all everyday forces (contact, friction, tension, normal) are **electromagnetic** at the atomic level.
- ▶ Gravity is extraordinarily weak but dominates at astronomical scales because it is always attractive and has infinite range.
- ▶ Electromagnetic + weak nuclear = **electroweak force** (unified at high energy).

Next Lecture

Chapter 5: Further Applications of Newton's Laws

- ▶ Static and kinetic friction — $f_k = \mu_k F_N$
- ▶ Drag force in more detail; Stokes' law
- ▶ Hooke's Law and elasticity: $\vec{F}_s = -k \Delta \vec{x}$
- ▶ Young's modulus and stress/strain

- ▶ The **four-step strategy** from today applies to all problems in Chapter 5.
- ▶ Review today's examples — inclined planes and connected systems appear frequently.